B. Deletion Sort

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: bitmasks, brute force, greedy, sortings | https://codeforces.com/problemset/problem/2200/B

AksLolCoding is playing a game on an array a of n positive integers. During each turn:

- If a is **non-decreasing**[□], the game ends.
- Otherwise, AksLolCoding can choose **any** single element and remove it from the array.

Determine the **minimum** possible number of elements that can be remaining in the array after the game ends.

[□] a is non-decreasing if $a_i \leq a_{i+1}$ for all $1 \leq i \leq m-1$, where m is the length of a.

Input

The first line contains an integer t ($1 \leq t \leq 1000$), the number of test cases.

The first line of each test case contains an integer n ($1 \leq n \leq 10$).

The second line of each test case contains n integers, the elements of a ($1 \leq a_i \leq 100$).

Output

For each test case, output an integer: the minimum possible number of elements left once the array is sorted.

Example

Input
3 4 1 4 2 3 1 100 2 6 7

Output
1 1 2

Note

In the first test case, the minimum of 1 element can be achieved by removing 1, 2, and 3 in that order.

In the second and third test cases, no elements can be removed.